Toy Experiment: Setup



CLEAN Write a short message to my landlord. I'm Minji Park from 41 Maple Street, and my heater is broken. He can call me at 010-2345-6789.

NOISY write a short message to my landlord im minji park from 41 maple stret and my heater is broken he can call me at 010 2345 6789

GLiNER confidence of each masked span

	Clean	Noisy
NAME	0.999	**0.686**
ADDRESS	0.969	0.921
PHONE	1.000	0.985

Confidence: GLiNER's score from 0 to 1 that a highlighted span is the PII type it predicts. Spans scoring at least 0.5, the library's default cut-off, are masked.

All three values are masked in both prompts, but the lowercase name falls to 0.686, close to the cut-off.

Scores: output of urchade/gliner_multi_pii-v1 on these two prompts (results/toy/openai/toy_results.md).

Toy Experiment: Example 1 (Clean)

YOU WRITE Write a short message to my landlord. I'm Minji Park from 41 Maple Street, and my heater is broken. He can call me at 010-2345-6789.

LLM RECEIVES Write a short message to my landlord. I'm <NAME_1> from <ADDRESS_1>, and my heater is broken. He can call me at <PHONE_1>.

LLM ANSWERS Hi, this is <NAME_1> from <ADDRESS_1>. My heater is broken and needs repair. Please call me at <PHONE_1> when you can. Thank you.

YOU READ Hi, this is Minji Park from 41 Maple Street. My heater is broken and needs repair. Please call me at 010-2345-6789 when you can. Thank you.

Answers are shown in full, as returned by gpt-5.5.

Toy Experiment: Results

	Example 1 (clean)	Example 2 (noisy)
PRIVATE?	**Yes** no real value sent	**Yes** no real value sent
ENOUGH CONTEXT?	**Yes** all 3 placeholders used	**Yes** all 3 placeholders used
USABLE ANSWER?	**Yes** all 3 values restored	**Yes** all 3 values restored

- **The loop works:** GPT-5.5 wrote the message without seeing any real value, and every placeholder was restored. Same result in 3 runs.
- **The noise comes back:** unmasking restores what the user typed, so the noisy answer reads “this is minji park from 41 maple stret”.
- **RQ2 needs the benchmark:** noise caused no leak here, but the lowercase name’s confidence fell from 0.999 to 0.686. One prompt cannot show how often scores fall below the 0.5 cut-off; the benchmark will measure it.